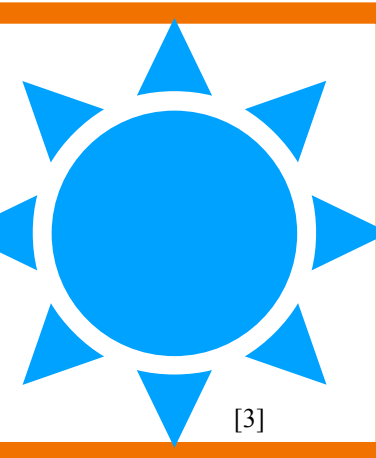
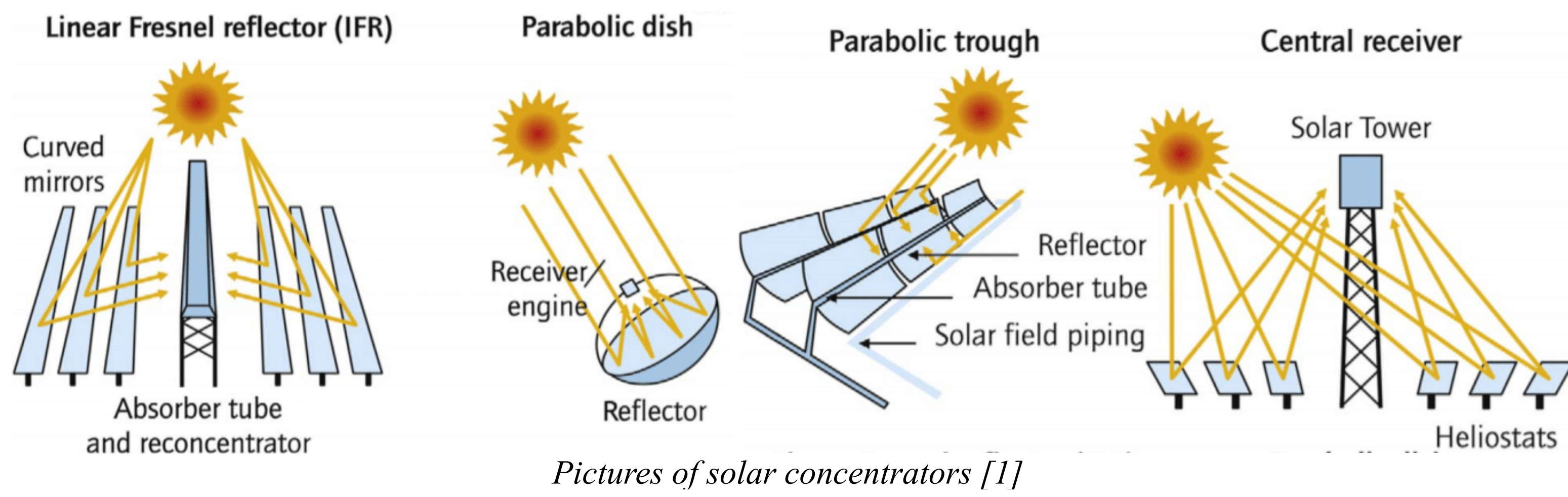




Solar energy can be harnessed effectively through the solar concentrators to meet electrical and thermal demand, and they can be classified into four categories, which are:



However, while there are many ideas to implement such solar systems in **sustainable** energy programs in ideal locations, like deserts, there is one common **enemy** that they all face: dust



Central Receiver in Gobi Desert [2]

Therefore, we decided to investigate and task ourselves to produce a different solar concentrator:
A Cylindrical Mirror system that enables the investigation of the power gain of a solar cell

Practical

Simulation

Engineering

Experiment

To measure intensity distribution of reflected light, a system with a detector on a rail was built.

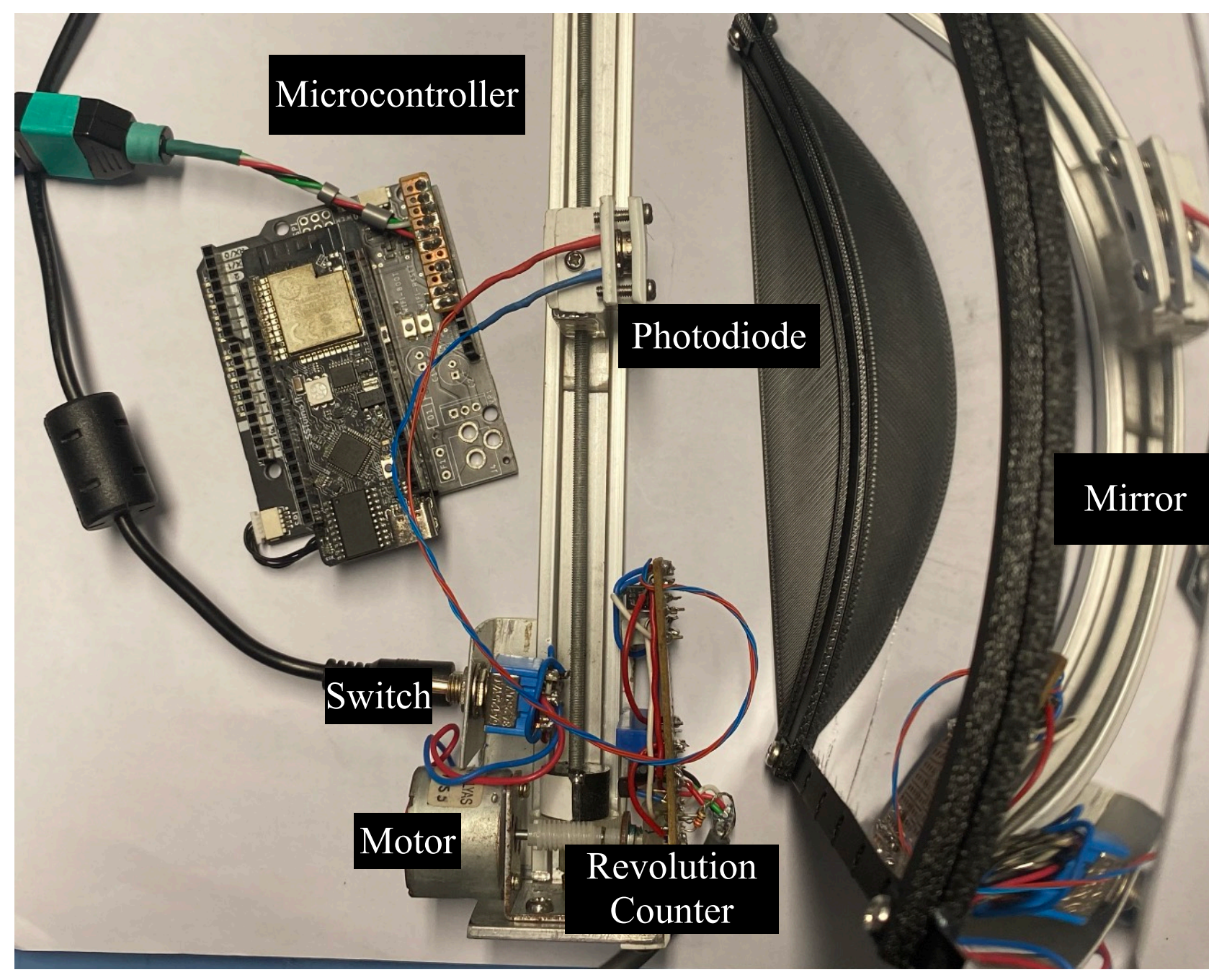
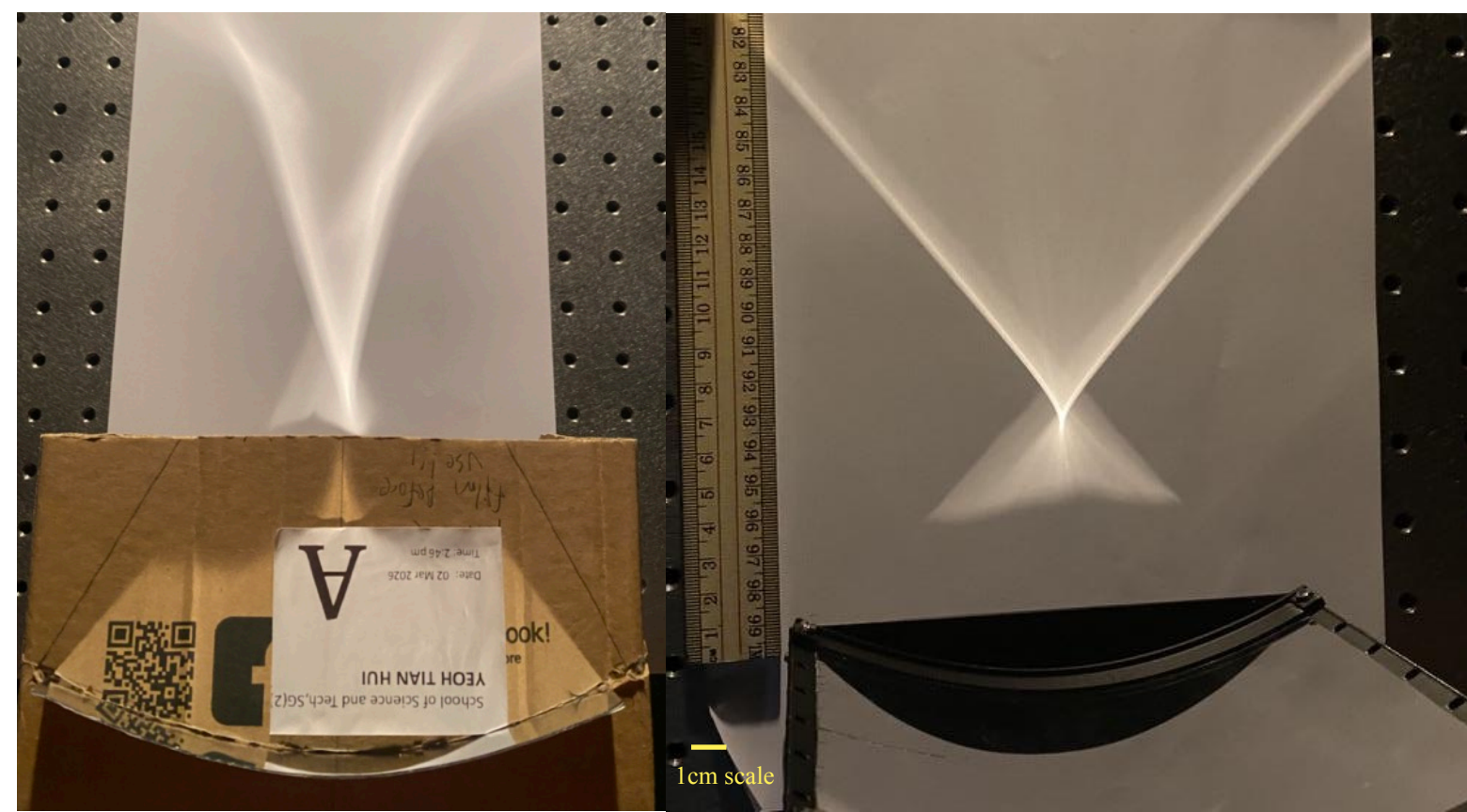


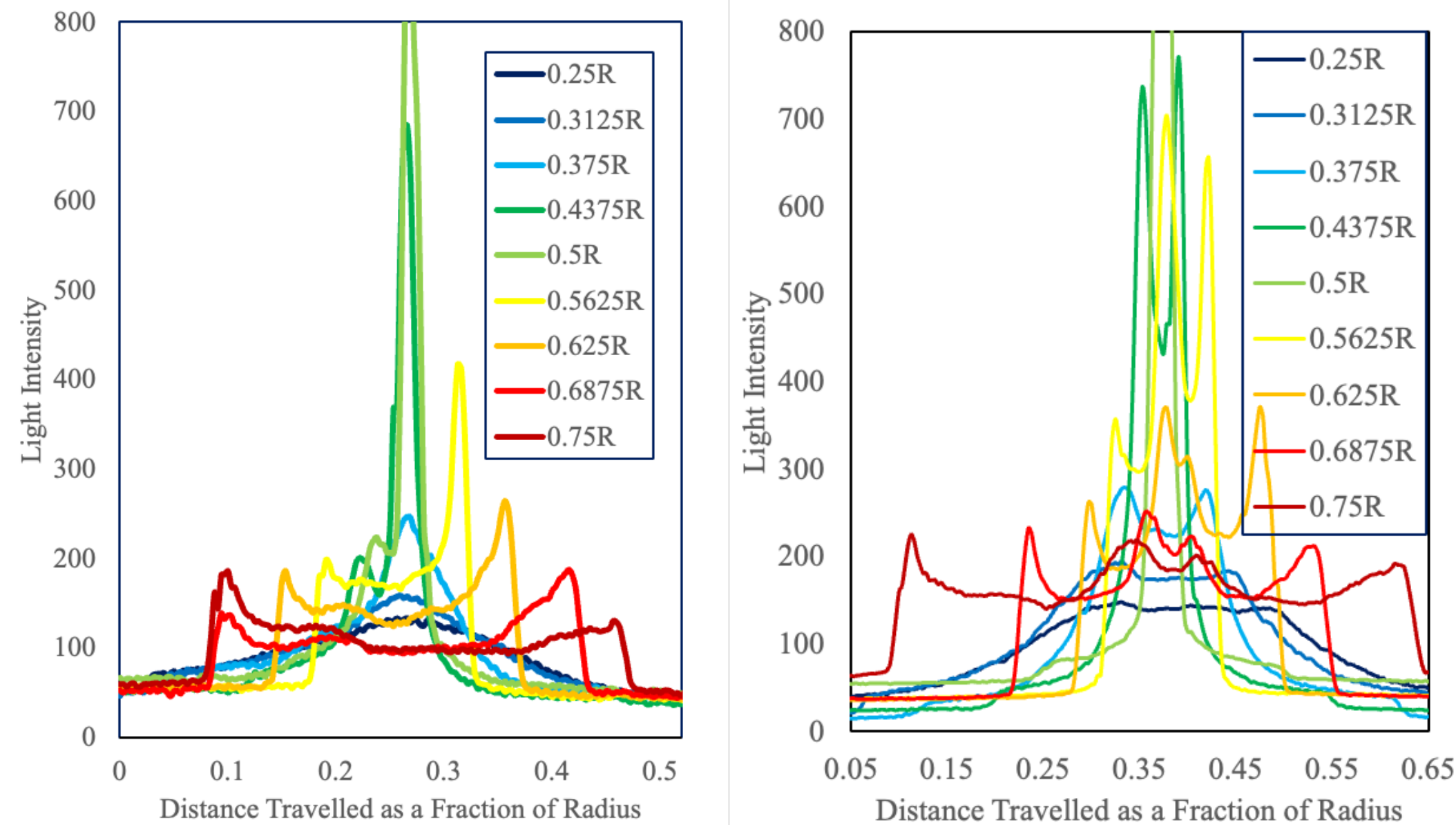
Photo of Set-up taken by the author

This system ensures that light intensity is measured at definite positions inside the mirror.



Top view of light projection from irregular (left) and regular (right) mirrors ($R=16\text{cm}$), photos taken by the author

Light Distribution



Properties shared between irregular and regular cylindrical mirrors.
Focal Area is located at a distance around $0.5R$.

Ray Tracing

To find the position X of a ray in measurement plane

$$R = 2 \cdot L \cos(\alpha)$$

$$L = \frac{R}{2 \cos(\alpha)}$$

$$X = (R - L - H) \cdot \tan 2\alpha$$

Incorporated into the Python Program:

H , the height to start step, height intervals
 b , number of divisions
 m , the maximum height rays, the number of rays

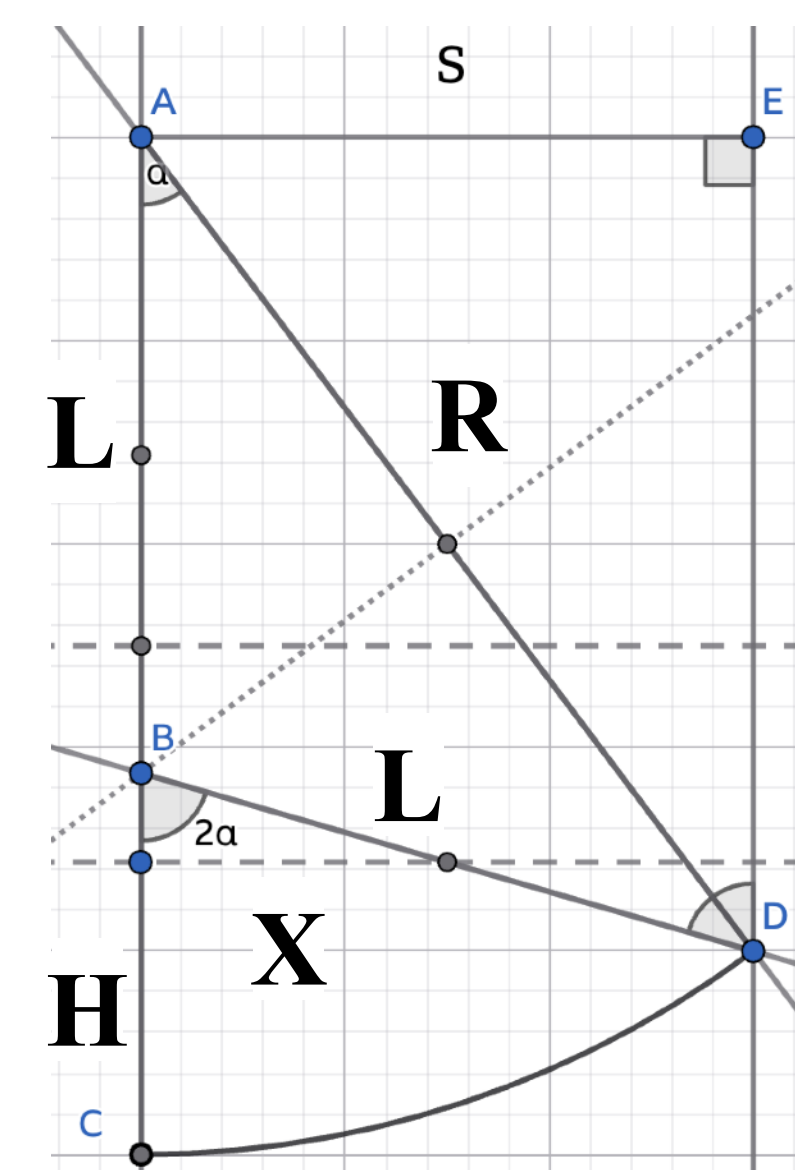
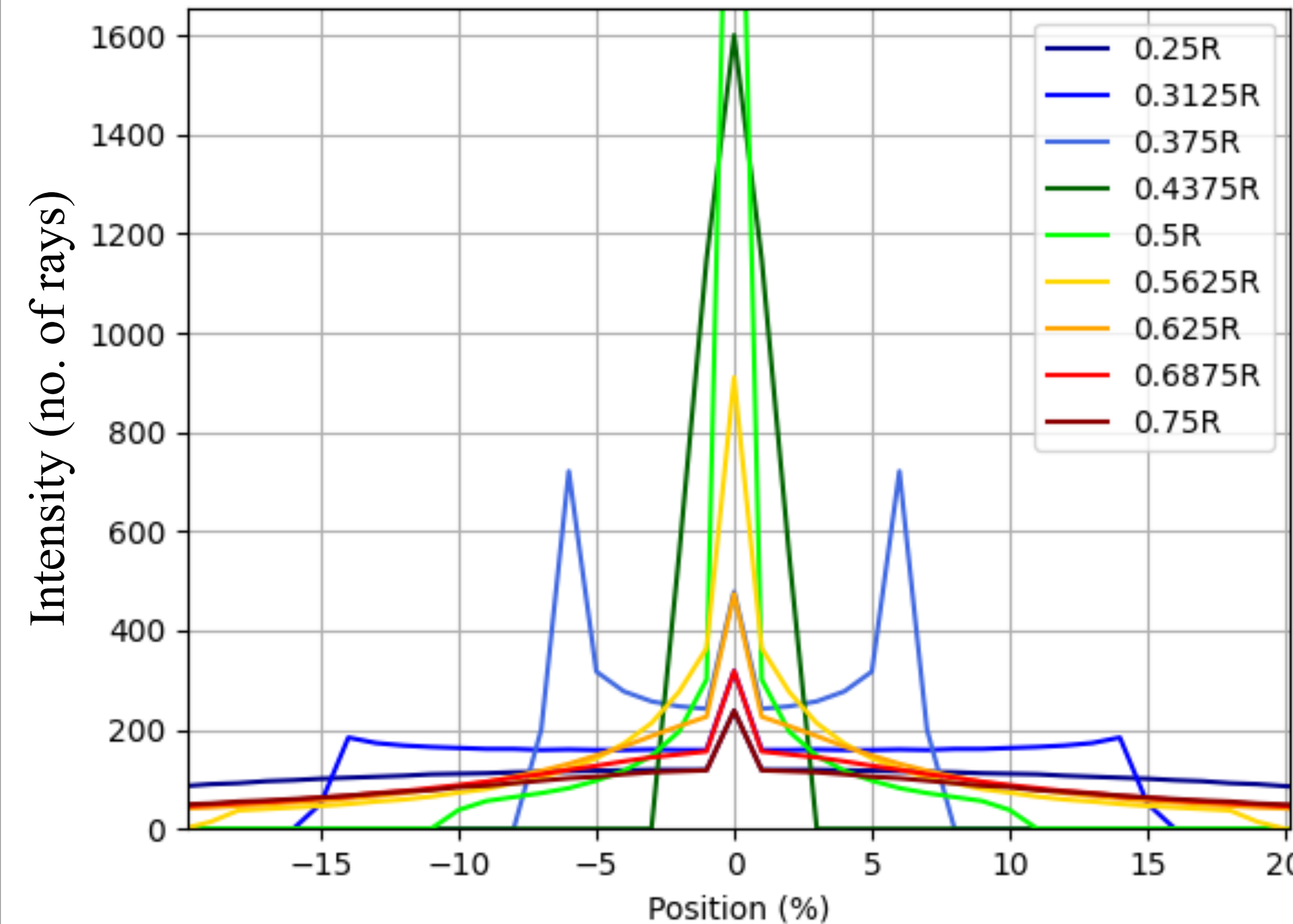


Diagram of a section of a theoretical cylindrical mirror
Plotted by the author on Geogebra



Intensity distribution calculated by ray counting from Python program, total number of rays is 5000, graphed by the author

Comparison

The shape of intensity distribution in experiment and model are similar. For quantitative comparison, we calculate integral width.

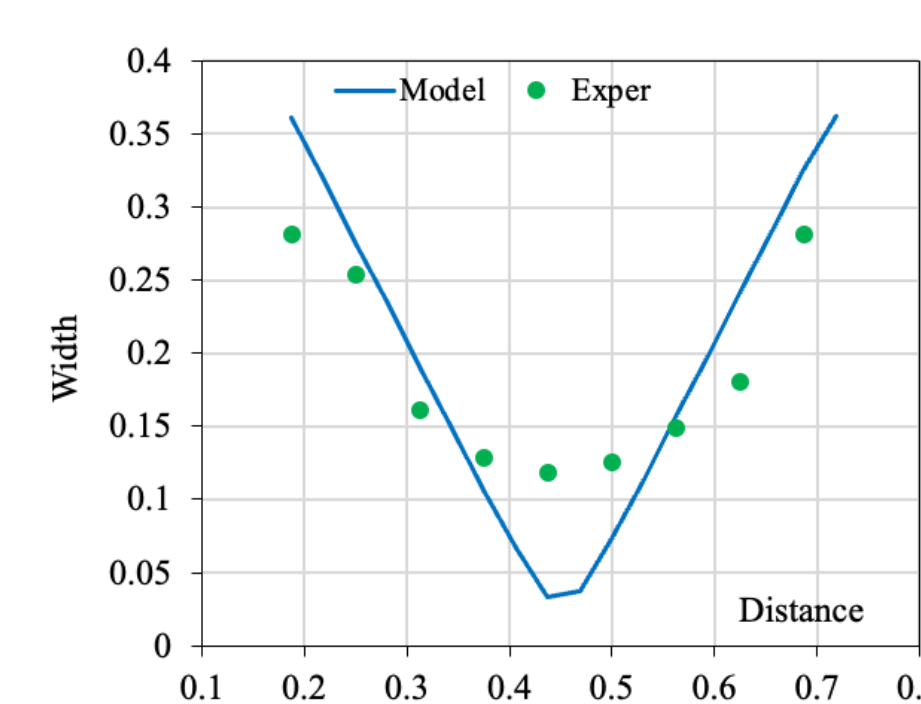
$$P = \frac{\sum_i y_i \cdot x_i}{\sum_i y_i}$$

Integral Position

$$W = \sqrt{\frac{3 \cdot \sum y_i \cdot (x_i - P)^2}{\sum y_i}}$$

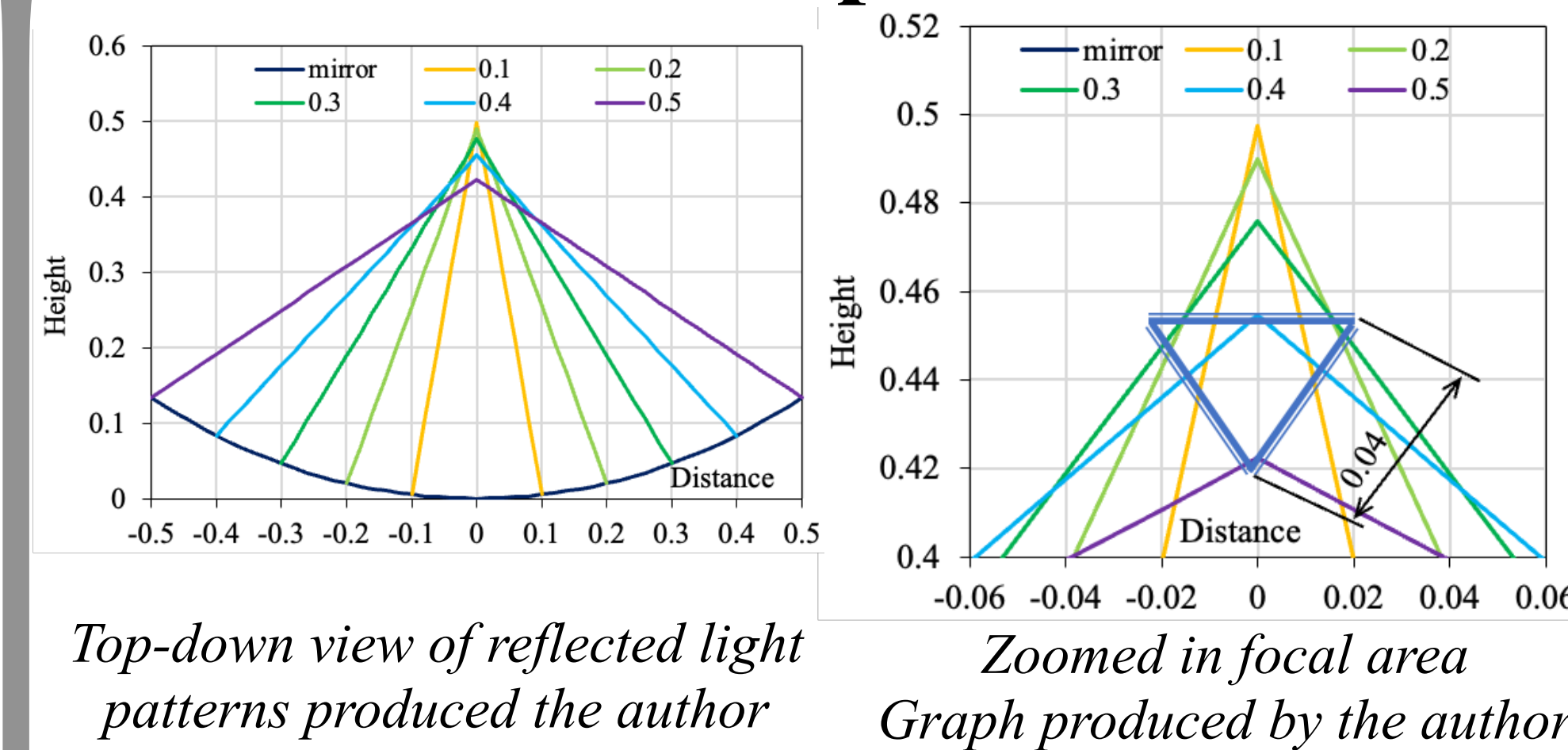
Integral Width

Focal Area is located at a distance of $0.45R$.



Integral width versus screen position created by the author

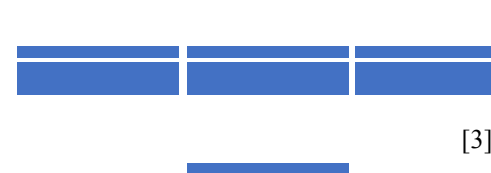
Development



Solar cells are assembled in an equilateral triangle to collect reflected and direct light. A total perimeter of around **12%** of the radius of a mirror is needed, so the theoretical power gain is **8** times.

Prototype

The geometrical limit of our gain is:
mirror size/panel size = $300\text{mm}/51\text{mm} \approx 6$ times

with V_1 when 
and V_2 when 

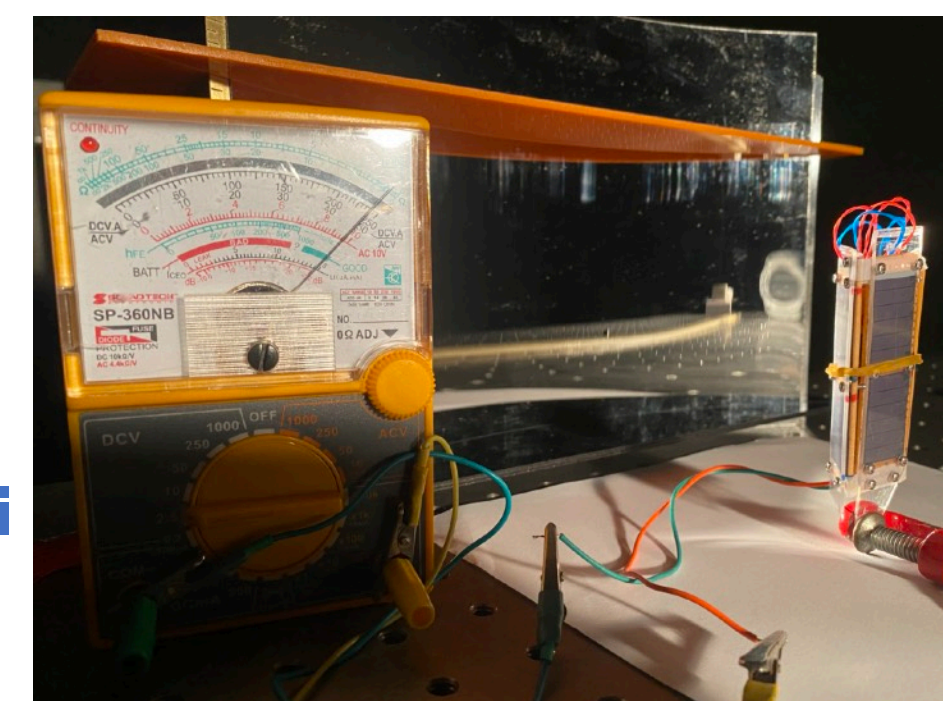
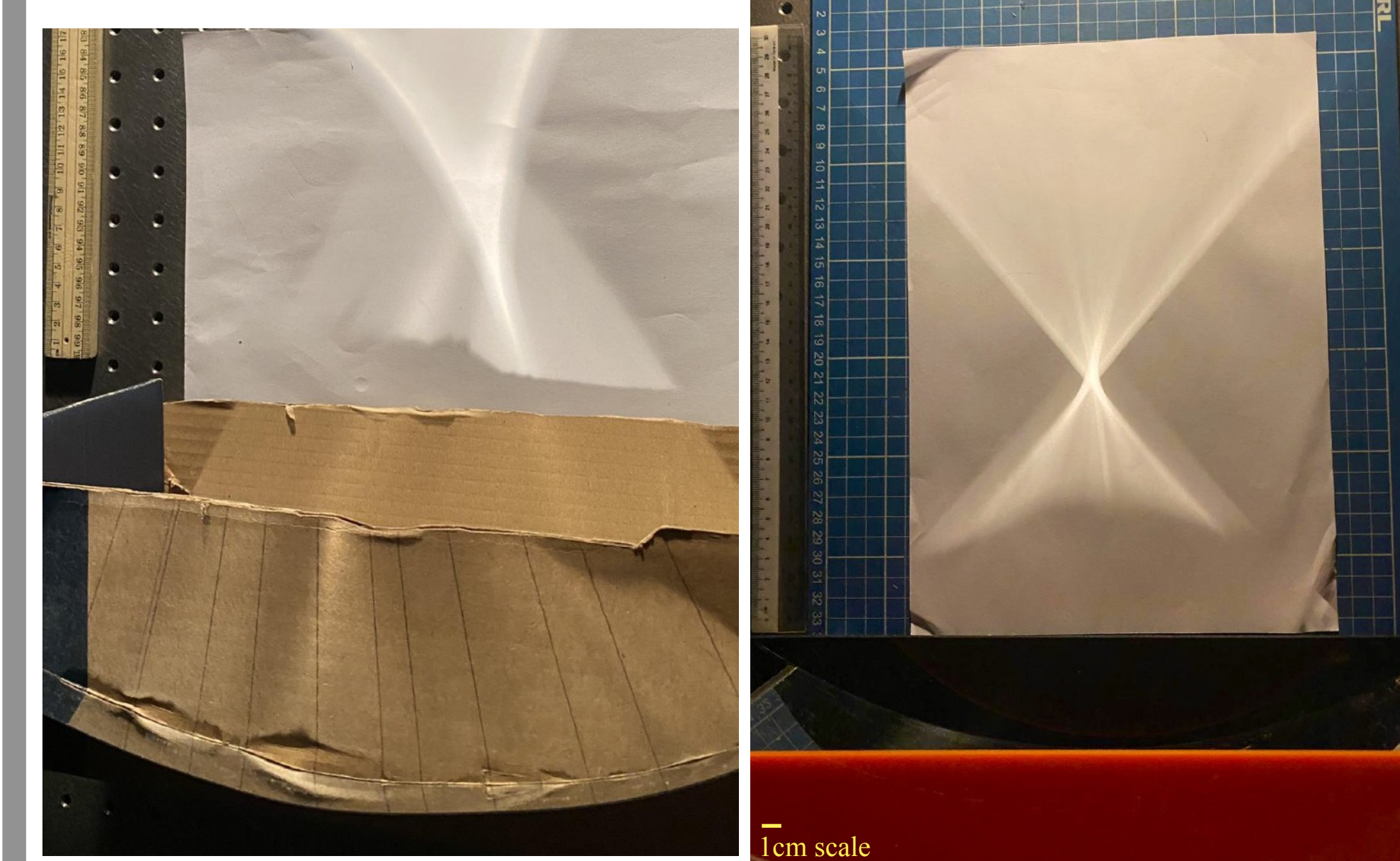


Photo of power measurement setup taken by the author

$$Power_{Gain} = \frac{P_2}{P_1} = \frac{V_2 I_2}{V_1 I_1} = \frac{\frac{V_2^2}{R}}{\frac{V_1^2}{R}} = \frac{V_2^2}{V_1^2}$$



Top view of light projection from irregular (left) and regular (right) mirrors ($R=30\text{cm}$), photos taken by the author

Power Gain from Mirror in Cardboard

$$\frac{2.1V^2}{1.05V^2} = 4.00$$

Power Gain from Regular Mirror

$$\frac{2.4V^2}{1V^2} = 5.76$$

Conclusion and Applications

- ✓ Physical and theoretical models share strong similarities
- 🔋 A system producing a power gain of 5.76 times is made
- ✂️ Cylindrical mirrors can be applied to practical setting

References

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- [2] Gandhi, S. (2025, October 9). World's first solar-thermal power plant commences in China's Gobi Desert. https://www.business-standard.com/world-news/china-solar-thermal-power-plant-gobi-desert-clean-energy-guazhou-125100900559_1.html
- [3] Icon taken from Pages
Infographic created by the author using Pages